Force

In physics, a **force** is a push or pull acting upon an object as a result of the object's interaction with another object.

It is an external agent capable of altering an object's state of rest or motion, causing it to accelerate, decelerate, or change direction.

Key Characteristics

- **Vector quantity** – Forces have both magnitude (strength) and direction.

- **SI unit** – The standard metric unit for force is the **Newton (N)**.

- One Newton is the force required to accelerate a 1 kg mass at a rate of **1 m s■^{2}**.

- **Formula** – Newton’s Second Law of Motion:

$$
\mathbf{F} = m\,\mathbf{a}
$$

where

- $\mathbf{F}$ = force,

- m = mass,

- $\mathbf{a}$ = acceleration.

Common Types of Forces

- **Contact forces** – Applied through physical contact between objects.

Examples: friction, tension, air resistance, and a physical push or pull.

- **Non■contact forces** – (not listed in the OCR but worth noting)

- Gravitational, electromagnetic, strong nuclear, and weak nuclear forces.

> **Note:** The OCR text contained several typographical errors (e.g., “Sl Unit” → “SI Unit”, “arate” → “rate”, “Fis” → “is”). These have been corrected for clarity.